ETH zürich

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Probabilistic Artificial Intelligence

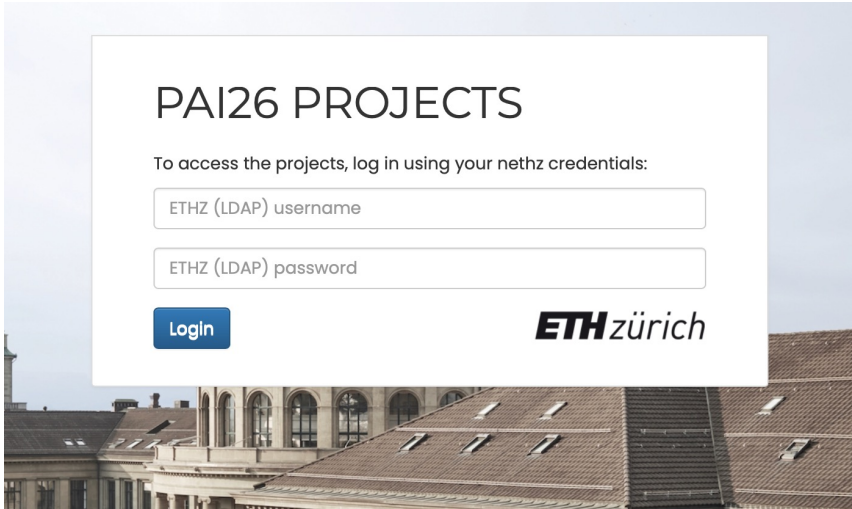
Bayesian Linear Regression

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Announcements

- Project part 0 is online: https://project.las.ethz.ch
- It is ungraded, but helps you become familiar with the workflow (solving it should take less than 30 min.)
- Submission remains open until Dec 31, but you're encouraged to complete it by Oct 2

- More info+tentative schedule on course webpage



Multivariate Gaussian distribution

$$N(x;\mu,\Sigma) = \frac{1}{(2\pi)^{n/2}\sqrt{|\Sigma|}}\exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$$

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1n} \\ \vdots & & & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_n^2 \end{pmatrix}$$

Thus joint distribution over *n* variables requires only $O(n^2)$ parameters!

Gaussians are closed under conditioning, marginalization, multiplication and addition

Conditional distributions

- Suppose we have a Gaussian random vector

$$\mathbf{X} = \mathbf{X}_V = [X_1, \ldots, X_d] \sim \mathcal{N}(\mu_V, \Sigma_{VV})$$

- Further, suppose we take two disjoint subsets of V

$$A = \{i_1, \ldots, i_k\} \qquad B = \{j_1, \ldots, j_m\}$$

- The conditional distribution

$$p(\mathbf{X}_A \mid \mathbf{X}_B = \mathbf{x}_B) = \mathcal{N}(\mu_{A|B}, \Sigma_{A|B})$$

is Gaussian, where

$$\mu_{A|B} = \mu_A + \Sigma_{AB}\Sigma_{BB}^{-1}(\mathbf{x}_B - \mu_B)$$
$$\Sigma_{A|B} = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}$$

Multiples of Gaussians are Gaussian

- Suppose we have a Gaussian random vector

$$\mathbf{X} = \mathbf{X}_V = [X_1, \ldots, X_d] \sim \mathcal{N}(\mu_V, \Sigma_{VV})$$

- Take a matrix $M \in \mathbb{R}^{m \times d}$
- Then the random vector $\mathbf{Y} = \mathbf{M}\mathbf{X}$ is Gaussian:

$$\mathbf{Y} \sim \mathcal{N}(\mathbf{M}\mu_V, \mathbf{M}\Sigma_{VV}\mathbf{M}^T)$$

Sums of Gaussians are Gaussian

- Suppose we have independent two Gaussian random vectors

$$\mathbf{X} = \mathbf{X}_V = [X_1, \ldots, X_d] \sim \mathcal{N}(\mu_V, \Sigma_{VV})$$
$$\mathbf{X}' = \mathbf{X}'_V = [X'_1, \ldots, X'_d] \sim \mathcal{N}(\mu'_V, \Sigma'_{VV})$$

- Then the random vector $\mathbf{Y} = \mathbf{X} + \mathbf{X}'$ is Gaussian:

$$\mathbf{Y} \sim \mathcal{N}(\mu_V + \mu'_V, \Sigma_{VV} + \Sigma'_{VV})$$

Observations: Conditional Linear Gaussians

- If X, Y are jointly Gaussian, then $p(X = x \mid Y = y)$ is Gaussian, with mean linearly dependent on *y*

$$p(X = x \mid Y = y) = N\left(x; \mu_{X|Y=y}, \sigma^2_{X|Y}\right)$$
$$\mu_{X|Y=y} = \mu_X + \sigma_{XY}\sigma_Y^{-2}(y - \mu_Y)$$
$$\sigma^2_{X|Y} = \sigma^2_X - \sigma^2_{XY}\sigma_Y^{-2}$$

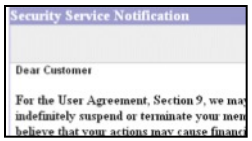
- Thus random variable X can be viewed as a linear function of Y with independent Gaussian noise added

$$X = aY + b + \epsilon, \text{ where } \epsilon \sim N(0, \sigma^2_{X|Y})$$
$$\text{and } a = \sigma_{XY}\sigma_Y^{-2}, b = \mu_X - \sigma_{XY}\sigma_Y^{-2}\mu_y$$

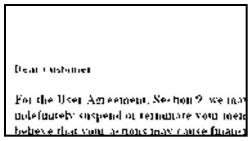
- The converse also holds.

Basic Supervised Learning Pipeline

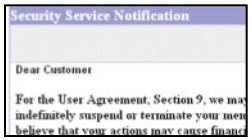
Training Data



"spam"



"ham"



"spam"

Learning method

Classi-fier

Predic-tion

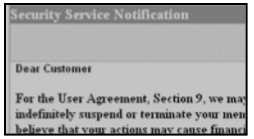
Test Data



?



?



?

$\mathcal{X}$ $\mathcal{Y}$

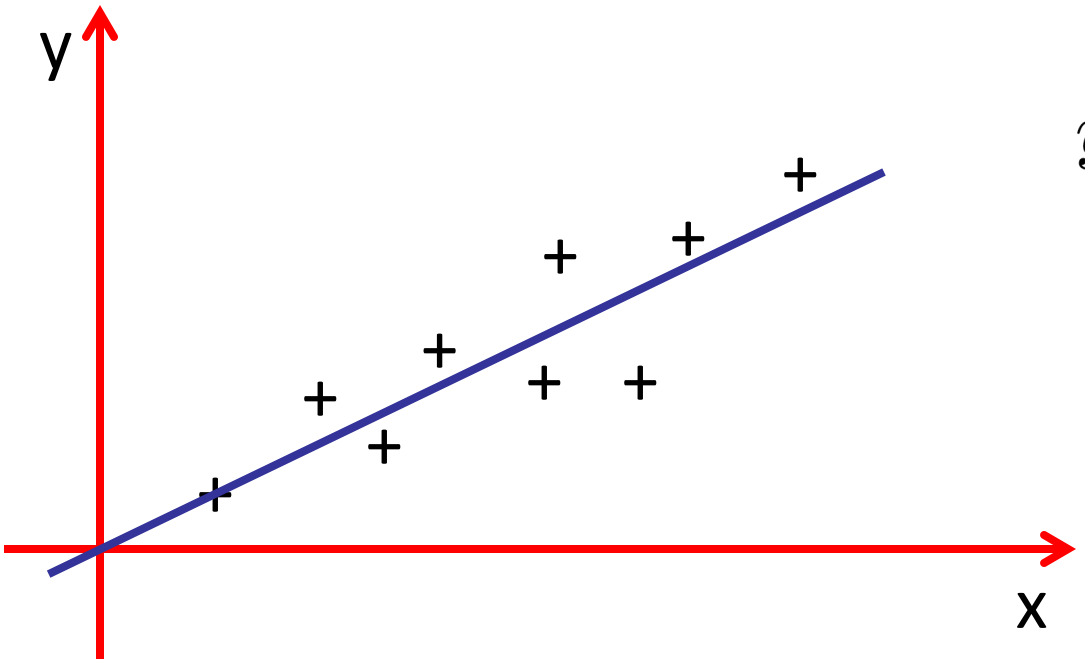
$f : \mathcal{X} \rightarrow \mathcal{Y}$

Model fitting

Prediction/ Generalization

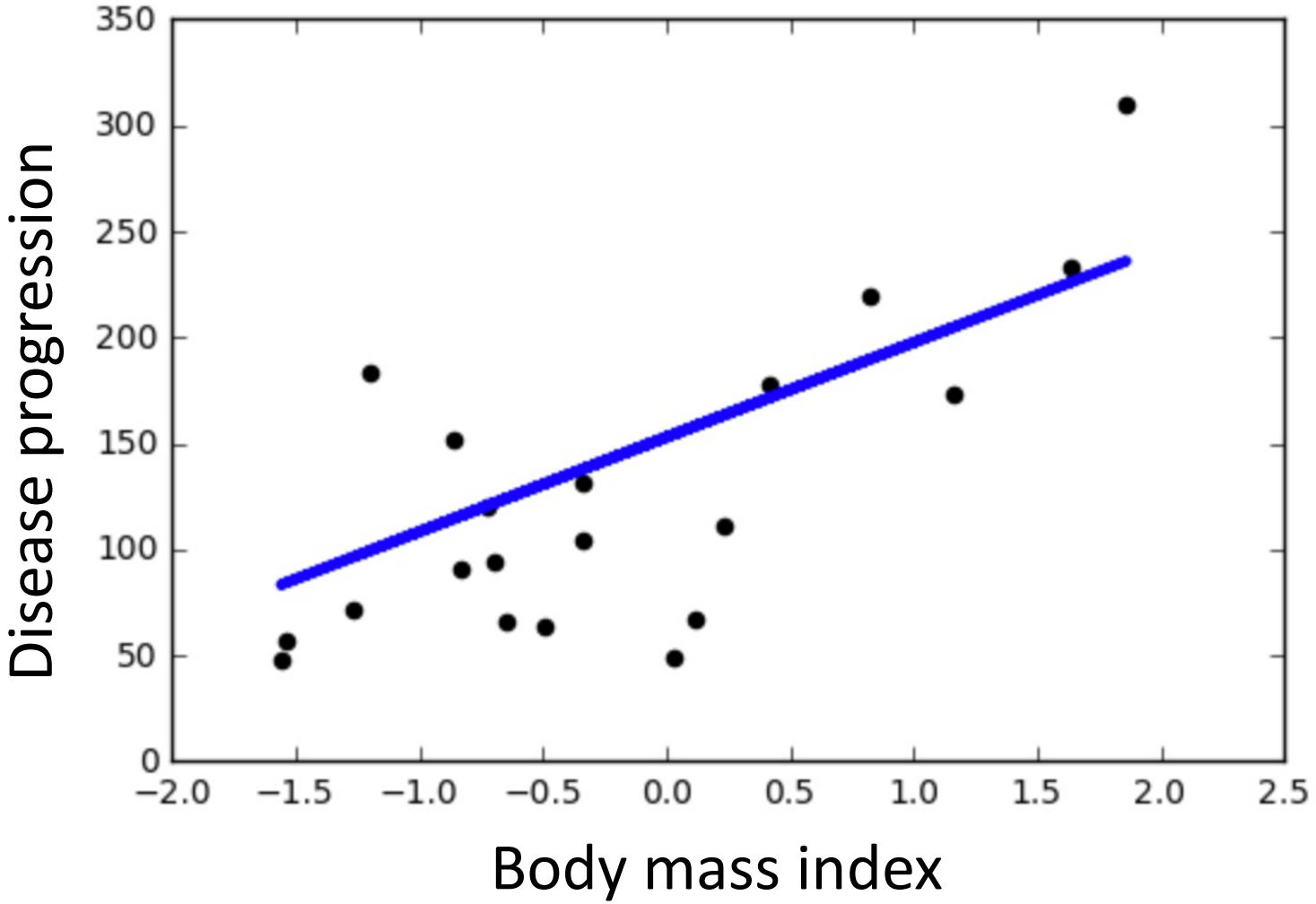
Bayesian Learning

Recall: linear regression



$$y \approx \mathbf{w}^T \mathbf{x}$$

Example


Disease progression
Body mass index
350
300
250
200
150
100
50
0
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5

Recall: Ridge regression

- Regularized optimization problem:

$$\min_{\mathbf{w}} \sum_{i=1}^{n} (y_i - \mathbf{w}^T \mathbf{x}_i)^2 + \lambda ||\mathbf{w}||_2^2$$

- Can optimize using (stochastic) gradient descent, or find analytical solution:

$$\hat{\mathbf{w}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

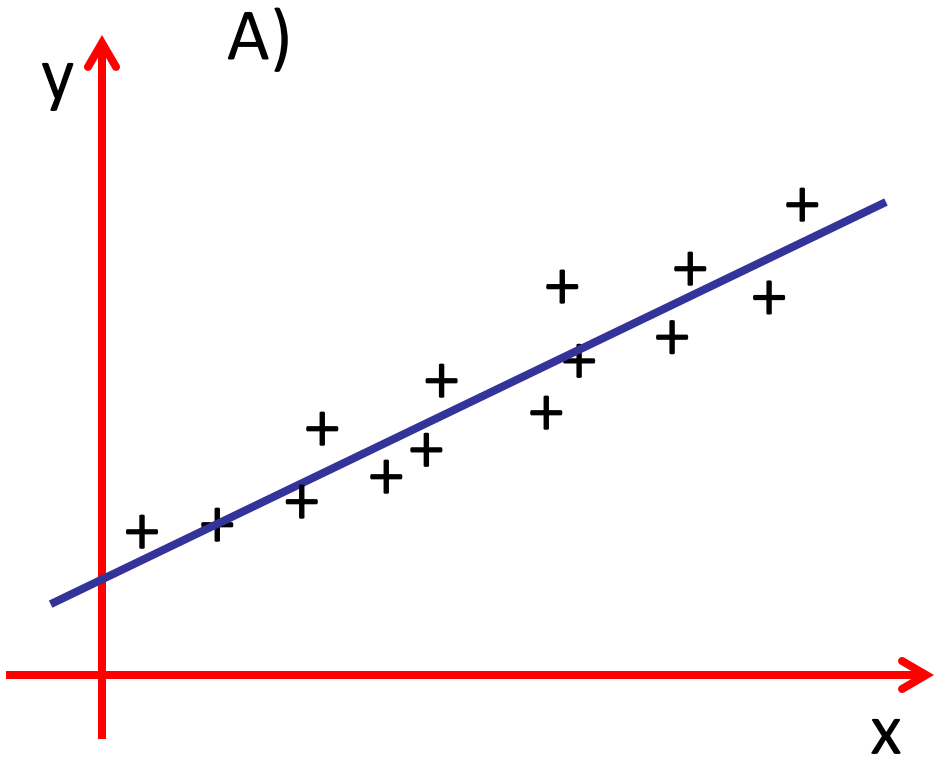
Ridge regression as Bayesian inference

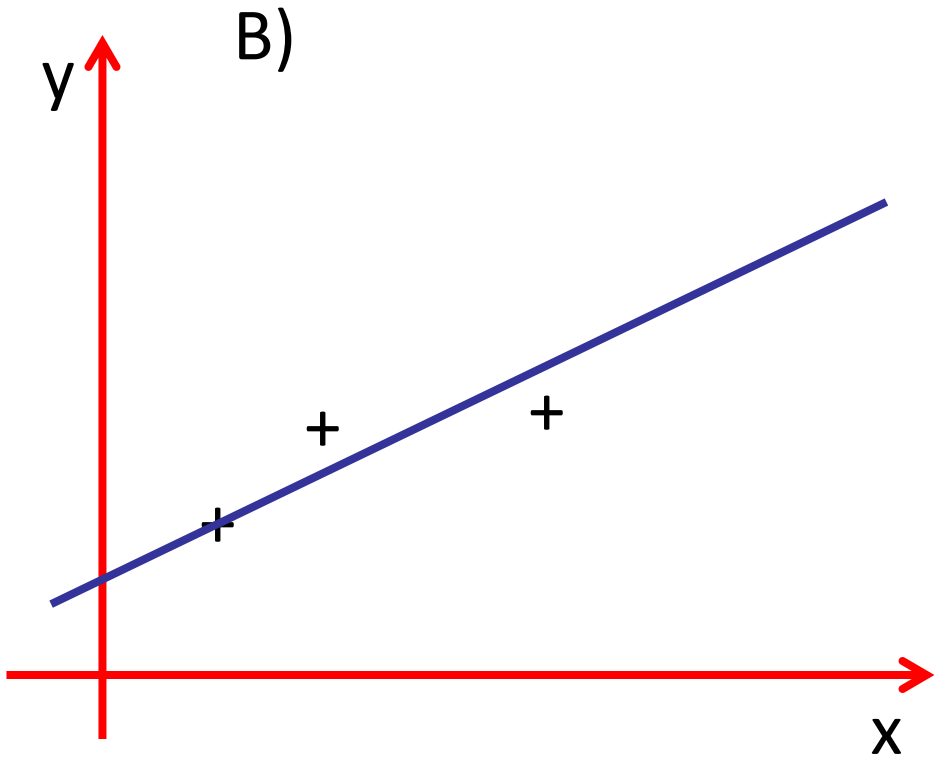
Ridge regression = MAP estimation

- Ridge regression can be understood as finding the Maximum A Posteriori (MAP) parameter estimate for a linear regression problem, assuming that
  - The noise *P(y|**x**,**w**)* is (cond.) iid Gaussian and
  - The prior *P(**w**)* on the model parameters ***w*** is Gaussian

$$\arg\min_{\mathbf{w}} \sum_{i=1}^{n} (y_i - \mathbf{w}^T \mathbf{x}_i)^2 + \lambda ||\mathbf{w}||_2^2 \equiv \arg\max_{\mathbf{w}} P(\mathbf{w}) \prod_i P(y_i \mid \mathbf{x}_i, \mathbf{w})$$

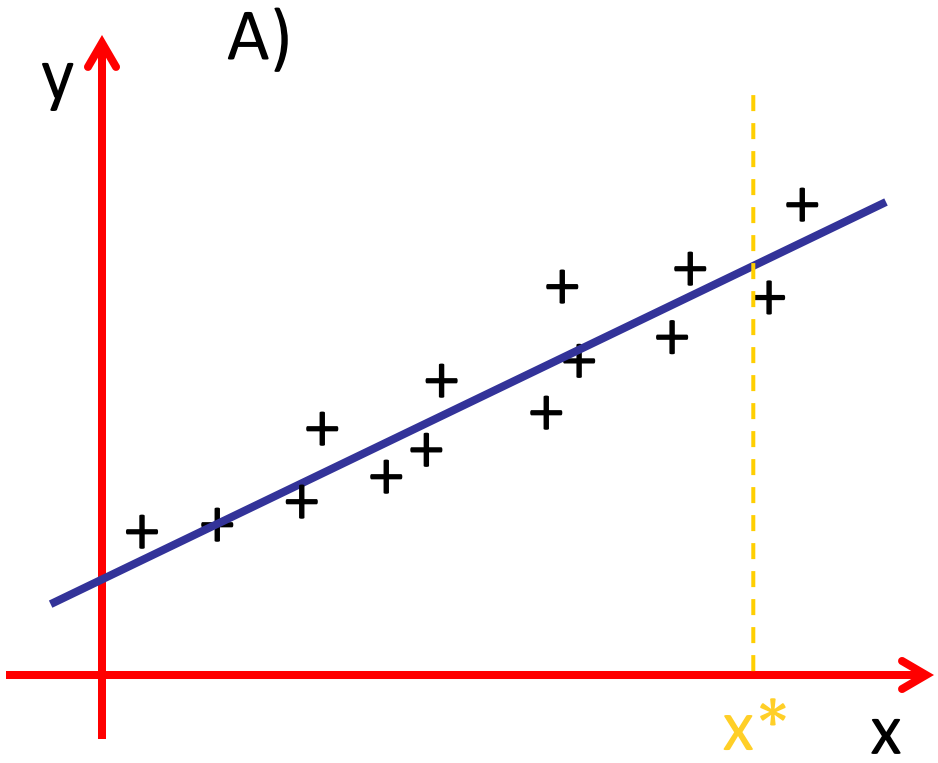
Model uncertainty

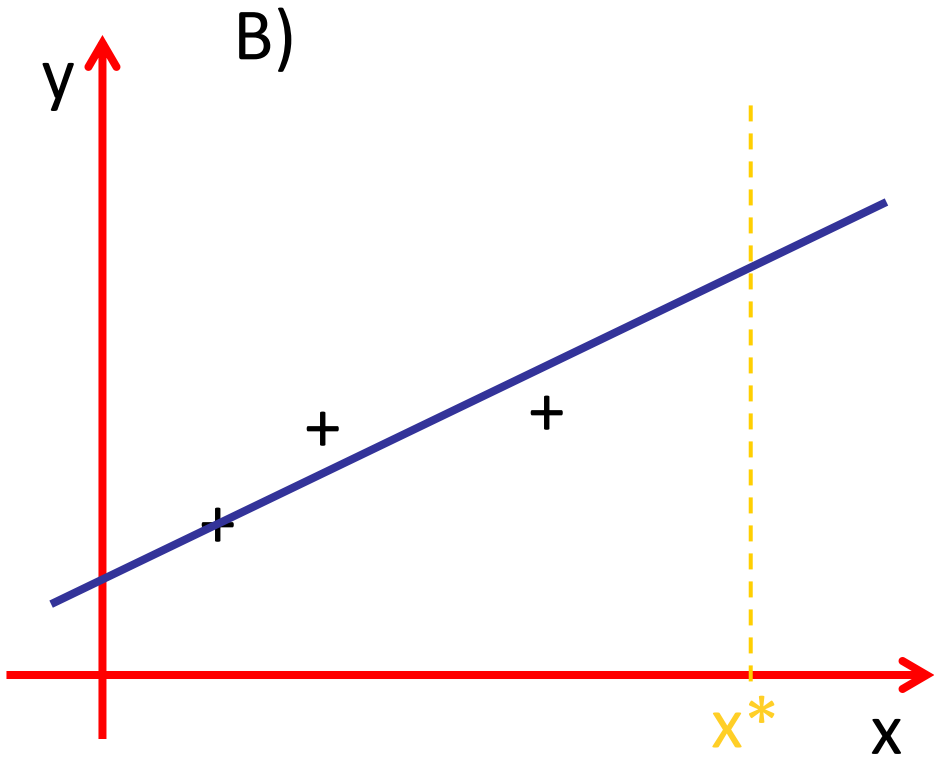




- Ridge regression returns a **single model**
- Such a point estimate does not quantify uncertainty

Making predictions





Bayesian Linear Regression (BLR)

- Key idea: Reason about full posterior of w, not only its mode

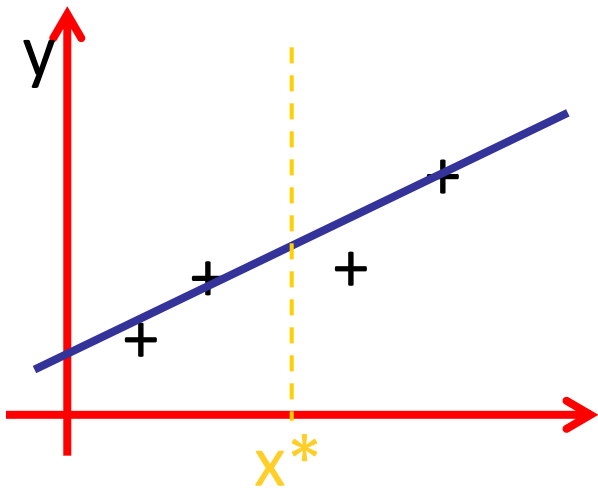
- For Bayesian linear regression with Gaussian prior and Gaussian likelihood, posterior has closed form

Posterior distributions in BLR

Prior: $p(\mathbf{w}) = \mathcal{N}(0, \mathbf{I})$

Likelihood:

$$p(y \mid \mathbf{x}, \mathbf{w}, \sigma_n) = \mathcal{N}(y; \mathbf{w}^T\mathbf{x}, \sigma_n^2)$$



Posterior: $p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \mathcal{N}(\mathbf{w}; \bar{\mu}, \bar{\Sigma})$

$$\bar{\mu} = (\mathbf{X}^T\mathbf{X} + \sigma_n^2\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$$

$$\bar{\Sigma} = (\sigma_n^{-2}\mathbf{X}^T\mathbf{X} + \mathbf{I})^{-1}$$

How can we use this to make predictions?

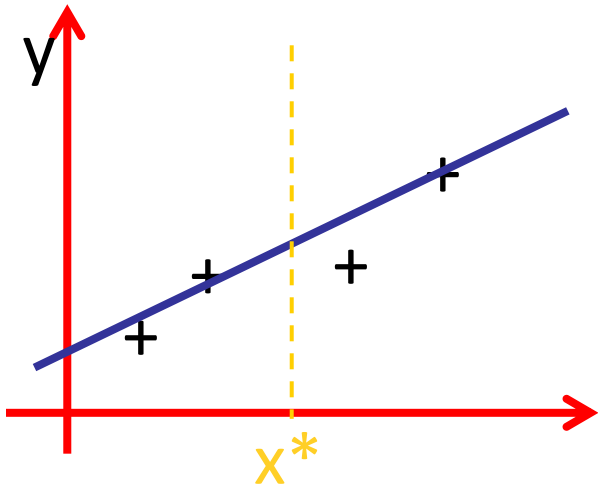
Making predictions in BLR

Posterior over **w**:

$$p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \mathcal{N}(\mathbf{w}; \bar{\mu}, \bar{\Sigma})$$

$$\bar{\mu} = (\mathbf{X}^T\mathbf{X} + \sigma_n^2\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$$

$$\bar{\Sigma} = (\sigma_n^{-2}\mathbf{X}^T\mathbf{X} + \mathbf{I})^{-1}$$



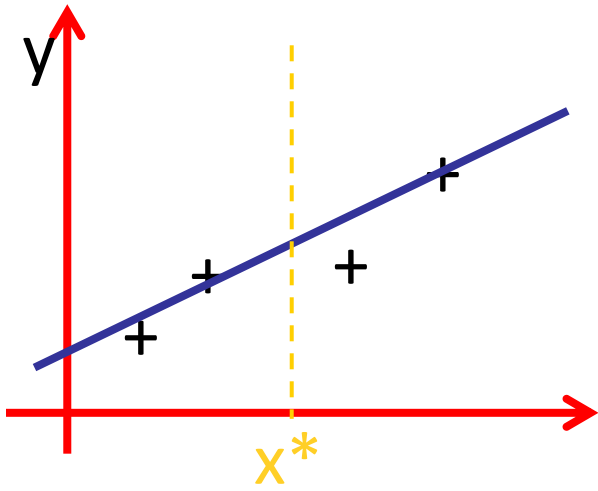
Making predictions in BLR

Posterior over **w**:

$$p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \mathcal{N}(\mathbf{w}; \bar{\mu}, \bar{\Sigma})$$

$$\bar{\mu} = (\mathbf{X}^T\mathbf{X} + \sigma_n^2\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$$

$$\bar{\Sigma} = (\sigma_n^{-2}\mathbf{X}^T\mathbf{X} + \mathbf{I})^{-1}$$



For test point $\mathbf{x}^*$ define $f^* = \mathbf{w}^T\mathbf{x}^*$. Then:

$$p(f^* \mid \mathbf{X}, \mathbf{y}, \mathbf{x}^*) = \mathcal{N}(\bar{\mu}^T\mathbf{x}^*, \mathbf{x}^{*T}\bar{\Sigma}\mathbf{x}^*)$$

and $$p(y^* \mid \mathbf{X}, \mathbf{y}, \mathbf{x}^*) = \mathcal{N}(\bar{\mu}^T\mathbf{x}^*, \mathbf{x}^{*T}\bar{\Sigma}\mathbf{x}^* + \sigma_n^2)$$

Ridge regression vs Bayesian lin. regression

- Ridge regression: predict using *MAP estimate* for weights

$$\hat{\mathbf{w}} = \arg\max_{\mathbf{w}} p(\mathbf{w} \mid \mathbf{x}_{1:n}, y_{1:n})$$

$$p(y^* \mid \mathbf{x}^*, \hat{\mathbf{w}}) = \mathcal{N}(y^*; \hat{\mathbf{w}}^T \mathbf{x}^*, \sigma_n^2)$$

- BLR: predict by averaging *all* **w** acc. to posterior:

$$p(y^* \mid x^*, x_{1:n}, y_{1:n}) = \int p(y^* \mid \boldsymbol{x}^*, \boldsymbol{w}) p(\boldsymbol{w} \mid \boldsymbol{x}_{1:n}, y_{1:n}) d\boldsymbol{w}$$

$$= N(y^*; \hat{w}^T \boldsymbol{x}^*, \sigma_n^2 + \boldsymbol{x}^{*T} \bar{\Sigma} \boldsymbol{x}^*)$$

Ridge regression vs Bayesian regression

- Thus, ridge regression can be viewed as approximating the full posterior by (placing all mass on) its mode

$$
\begin{aligned}
p(y^* \mid \mathbf{x}^*, \mathbf{x}_{1:n}, y_{1:n}) &= \int p(y^* \mid \mathbf{x}^*, \mathbf{w}) p(\mathbf{w} \mid \mathbf{x}_{1:n}, y_{1:n}) d\mathbf{w} \\
&\approx \int p(y^* \mid \mathbf{x}^*, \mathbf{w}) \delta_{\hat{\mathbf{w}}}(\mathbf{w}) d\mathbf{w} \\
&= p(y^* \mid \mathbf{x}^*, \hat{\mathbf{w}})
\end{aligned}
$$

$$
\hat{\mathbf{w}} = \arg\max_{\mathbf{w}} p(\mathbf{w} \mid \mathbf{x}_{1:n}, y_{1:n})
$$

Aleatoric vs. epistemic uncertainty

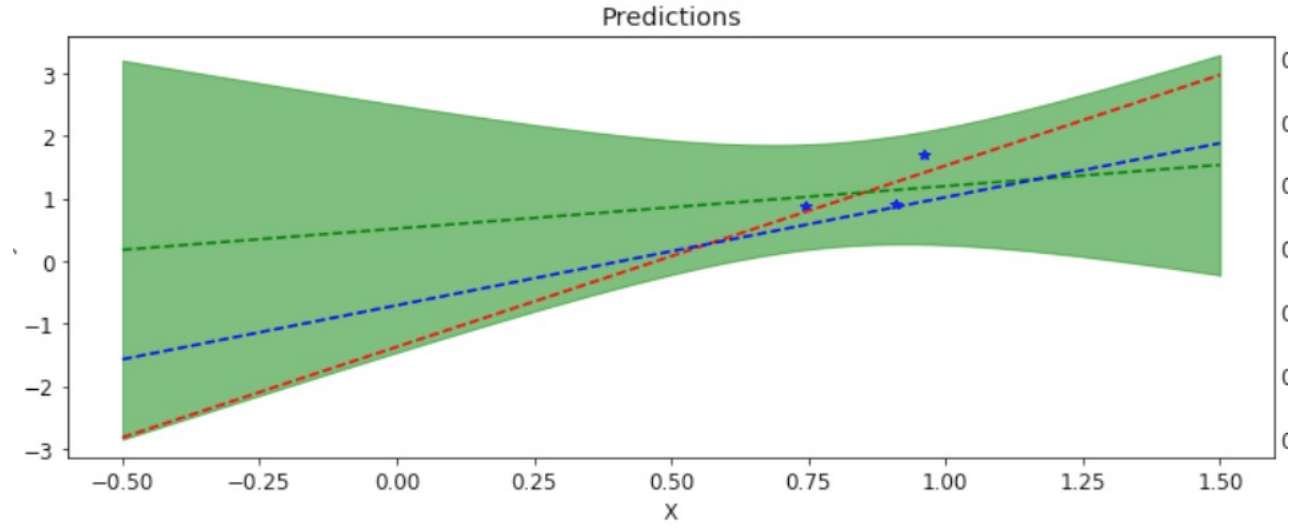
$$p(y^* \mid \mathbf{X}, \mathbf{y}, \mathbf{x}^*) = \mathcal{N}(\bar{\mu}^T \mathbf{x}^*, \mathbf{x}^{*T} \bar{\Sigma} \mathbf{x}^* + \sigma_n^2)$$

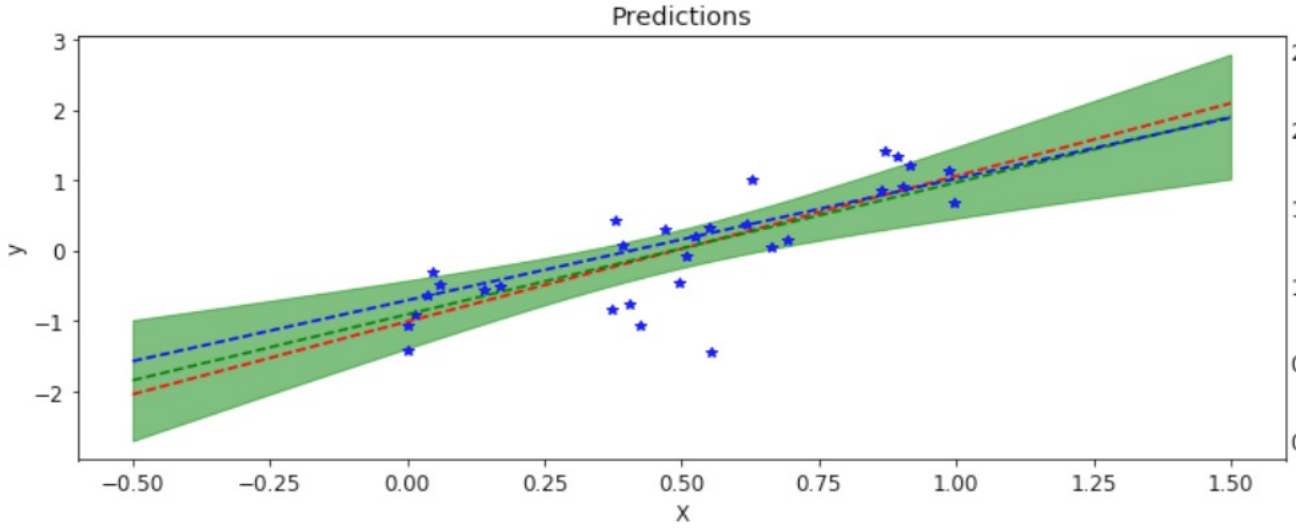
Uncertainty about f* (epistemic)

Noise / uncertainty about y* given f* (aleatoric)

- Can distinguish two forms of uncertainty
  - Epistemic uncertainty: Uncertainty about the model due to the lack of data
  - Aleatoric uncertainty: Irreducible noise

Demo: BLR


Predictions
3
2
1
0
−1
−2
−3
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
X


Predictions
y
3
2
1
0
−1
−2
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
X

Bayesian decision theory

- Given:
  - Conditional distribution over labels $P(y \mid x), y \in \mathcal{Y}$
  - Set of actions (decisions): $\mathcal{A}$
  - Cost function: $C: \mathcal{Y} \times \mathcal{A} \rightarrow \mathbb{R}$
- Bayesian Decision Theory recommends to pick the action that minimizes the expected cost

$$a^* = \arg\min_{a \in \mathcal{A}} \mathbb{E}_y[C(y, a) \mid \mathbf{x}]$$

Optimal decisions for squared loss

- Est. cond. dist.: $P(y \mid x) = \mathcal{N}(y; \mu_{Y|X=x}, \sigma^2_{Y|X})$
- Action set: $\mathcal{A} = \mathbb{R}$
- Cost function: $C(y, a) = (y - a)^2$

- Then the action that minimizes the expected cost

$$a^* = \arg\min_{a \in \mathcal{A}} \mathbb{E}_y[C(y, a) \mid \mathbf{x}]$$

is the conditional mean:

$$a^* = \mathbb{E}_y[Y \mid x] = \int y \, p(y \mid x) dx$$
$$= \mu_{Y|X=x}$$

Asymmetric cost for regression

- Est. cond. dist.: $P(y \mid x) = \mathcal{N}(y; \mu_{Y|X=x}, \sigma^2_{Y|X})$
- Action set: $\mathcal{A} = \mathbb{R}$
- Cost: $C(y, a) = c_1 \max(y - a, 0) + c_2 \max(a - y, 0)$
-

- Then the action that minimizes the expected cost

$$a^* = \arg\min_{a \in \mathcal{A}} \mathbb{E}_y[C(y, a) \mid \mathbf{x}]$$

is: $$a^* = \mu_{Y|X=x} + \sigma_{Y|X} \Phi^{-1}\left(\frac{c_1}{c_1 + c_2}\right)$$

Choosing hyperparameters

- In BLR, need to specify the (co-)variance of the prior $\boldsymbol{\sigma}_p$ and the variance of the noise $\boldsymbol{\sigma}_n$
- These are hyperparameters of the model (governing the distribution of the parameters $\boldsymbol{w}$)
- How should we choose them? One option:
  - Choose $\hat{\lambda} = \frac{\hat{\sigma}_n^2}{\hat{\sigma}_p^2}$ via cross-validation (see IntroML)
  - Then estimate $\hat{\sigma}_n^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{w}^T \mathbf{x}_i)^2$ as the empirical variance of the residual, and solve for $\hat{\sigma}_p^2 = \frac{\hat{\sigma}_n^2}{\hat{\lambda}}$
- In later lectures, we'll learn about a different way, considering the marginal likelihood of the data

Side note: Graphical models

- Have seen: Can represent arbitrary joint distributions as product of conditionals via chain rule

- Often, factors only depend on subsets of variables

- Can represent the resulting product as a directed acyclic graph

Graphical model for BLR

Recursive Bayesian updates

- “Today’s posterior is tomorrow’s prior”

Summary Bayesian Linear Regression

- Bayesian linear regression makes same modeling assumptions as ridge regression
  (Gaussian prior on weights, Gaussian noise)
- BLR computes / uses full posterior distribution over the weights rather than the mode only
- Thus, it captures uncertainty in weights, and allows to separate epistemic from aleatoric uncertainty
- Due to independence of the noise, can do recursive updates on the weights

Reading

- Lecture notes: Chapter 2

- Deisenroth, Faisal & Ong: Mathematics for ML
  - Chapter 6.5
  - Chapter 9
- Rasmussen & Williams: Gaussian Processes for ML
  - Chapter 2 up to 2.1.1

